

FINAL YEAR PROJECT

LOGBOOK

FOR

VIII SEMESTER B.Tech. (ISE)

Title of the Project

**A Self-Learning Image Quality Assessment and Enhancement Pipeline
for Autonomous Drone Photogrammetry**

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Conclusion

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 Signature of the Project Coordinator

 Signature of the HOD

Weekly Project Log Report

Week: Week 1

Date: _____

Task Completed

- Identified the core problem statement: automated image quality assessment for drone photogrammetry.
- Conducted brainstorming sessions on UAV image degradation types: blur, haze, noise, low-light.
- Collected information on relevant technologies (CNNs, EfficientNet, IQA metrics, GMM clustering, enhancement pipelines).
- Prepared the project synopsis document summarizing objectives, requirements, and feasibility.
- Finalized the scope, proposed outcomes (GOOD/RECOVERABLE/BAD classification), and high-level system workflow.

Task in Progress

- Continuing to refine understanding of problem complexity and no-reference IQA approaches.
- Exploring similar existing systems to identify research gaps.

Task to Begin Next Week

- Begin an extensive literature survey.
- Identify key research papers related to image quality assessment, UAV imagery, and deep learning enhancement.

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Weekly Project Log Report

Week: Week 2 & 3

Date: _____

Task Completed

- Reviewed approximately 20+ research papers related to: No-reference IQA, UAV/drone image enhancement, ensemble deep learning, self-supervised methods.
- Extracted key aspects from each paper: problem addressed, dataset used, methodology, key contributions, merits and limitations.
- Identified potential models/technologies (EfficientNet, BRISQUE, NIMA, PIQA, GMM, Retinex, DCP, Richardson-Lucy).
- Analyzed possible research gaps in unified self-learning pipelines for drone imagery.
- Created a structured PPT summarizing all reviewed papers for guide review.

Task in Progress

- Shortlisting the most relevant model architectures for experimentation.
- Discussing innovative features such as dual-path labeling and self-learning adaptation.

Task to Begin Next Week

- Begin collecting drone imagery dataset.
- Explore public dataset sources (Kaggle, Roboflow) and finalize quality categories.

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Weekly Project Log Report

Week: Week 4

Date: _____

Task Completed

- Collected drone images from publicly available sources (Kaggle, online UAV datasets) and open repositories.
- Generated synthetic degraded images (haze, blur, noise) using image processing tools for BAD class augmentation.
- Performed manual filtering to ensure representative spread across GOOD, RECOVERABLE, and BAD categories.
- Finalized a curated collection of 900+ drone images covering rural, agricultural, and terrain scenarios.
- Organized images into structured folders for annotation and preprocessing.

Task in Progress

- Removing duplicates and low-quality images.
- Verifying coverage across different lighting conditions, flight altitudes, and degradation types.

Task to Begin Next Week

- Begin annotation using metric-based thresholding.
- Finalize dataset structure and export formats for model training.

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Weekly Project Log Report

Week: Week 5 & 6

Date: _____

Task Completed

- Computed IQA metrics on all images: Laplacian variance, Tenengrad gradient, ORB keypoints, edge density, luminance, SNR, haze score.
- Applied z-score normalization and quartile-based multi-criteria rules to assign GOOD, RECOVERABLE, BAD labels.
- Generated synthetic distortions (Gaussian blur, noise, artificial haze) to augment the BAD class.
- Performed preprocessing: resizing, RGB normalization, histogram equalization, and optional perspective correction.
- Exported dataset in structured format for training the EfficientNet-B0 classifier.

Task in Progress

- Validating annotation consistency across metric thresholds.
- Creating training/validation/test splits with balanced class distribution.

Task to Begin Next Week

- Begin Phase 1 documentation and prepare slides summarizing work completed so far.
- Start implementing the Deep-IQA driven clustering path using PIQA and NIMA models.

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Weekly Project Log Report

Week: Week 7

Date: _____

Task Completed

- Prepared a complete Phase 1 PowerPoint including: problem overview, literature survey summary, proposed system workflow, dataset collection methodology, and dual-path labeling approach.
- Finalized a project workflow diagram illustrating: image input, dual-path labeling, EfficientNet-B0 classification, hybrid enhancement, re-classification, and self-learning loop.
- Presented Phase 1 deliverables to the evaluation panel.

Task in Progress

- Addressing evaluator feedback on pipeline design.
- Improving dataset balance and annotation quality based on suggestions received.

Task to Begin Next Week

- Begin implementing the Deep-IQA-driven clustering path (PIQA, NIMA, GMM).
- Start model training for EfficientNet-B0 classifier.

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Weekly Project Log Report

Week: Week 8 & 9

Date: _____

Task Completed

- Implemented the Deep-IQA-Driven Clustering path: computed perceptual scores using PIQA, NIMA, and CNN-based predictors.
- Fused perceptual scores into a composite quality index using weighted combination.
- Fitted a 3-component Gaussian Mixture Model (GMM) on feature vectors; assigned GOOD, RECOVERABLE, BAD labels by maximum posterior probability.
- Implemented label fusion logic: consistent labels merged directly; conflicting labels defaulted to RECOVERABLE.
- Saved labeled datasets with metadata for full reproducibility.

Task in Progress

- Evaluating agreement between metric-based and deep-IQA paths across all images.
- Analyzing edge cases where labeling conflict is highest.

Task to Begin Next Week

- Begin EfficientNet-B0 classifier training on the fused labeled dataset.

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Weekly Project Log Report

Week: Week 10

Date: _____

Task Completed

- Trained EfficientNet-B0 classifier (head replaced with 3-logit FC layer) on fused labeled dataset.
- Used categorical cross-entropy loss with Adam optimizer and cosine learning rate decay.
- Trained with FP16 mixed precision and data-parallel distribution on GPU.
- Evaluated model on validation set; achieved initial validation accuracy around 78-82%.
- Analyzed confusion matrix and per-class precision/recall/F1 metrics.

Task in Progress

- Improving RECOVERABLE vs BAD discrimination through hyperparameter tuning.
- Experimenting with augmentations to boost model generalization.

Task to Begin Next Week

- Implement and test the Multi-Stage Quality Refinement Pipeline.
- Begin development of the hybrid image enhancement module.

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Weekly Project Log Report

Week: Week 11 & 12

Date: _____

Task Completed

- Designed and implemented the four-stage quality refinement pipeline: Initial Prediction, Hybrid Enhancement, Re-Classification, and Logging.
- Stage 1 routes GOOD images directly to photogrammetry and passes others to enhancement.
- Stage 3 re-classifies enhanced images; promotes to GOOD if improved, discards persistent BAD.
- Stage 4 logs initial/final labels and enhancement outcomes for monitoring and self-learning.
- Validated pipeline end-to-end on held-out drone image sequences.

Task in Progress

- Fine-tuning threshold values for stage transitions.
- Testing pipeline performance on images with multiple simultaneous degradation types.

Task to Begin Next Week

- Implement the hybrid enhancement module (dehazing, denoising, deblurring).

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Weekly Project Log Report

Week: Week 13

Date: _____

Task Completed

- Implemented the hybrid enhancement module: DCP-based dehazing using integral images for efficiency.
- Added GPU bilateral filtering for edge-preserving denoising.
- Integrated FFT-accelerated Richardson-Lucy deblurring for sharpness restoration.
- Combined pipeline follows: dehazing → denoising → deblurring order.
- Achieved processing time of 100–150 ms per frame on mid-tier GPU.

Task in Progress

- Tuning dehazing transmission map estimation for aerial scenes.
- Evaluating enhancement quality using PSNR and SSIM metrics on recoverable images.

Task to Begin Next Week

- Implement the self-learning adaptation mechanism.
- Design the dynamic dataset and periodic fine-tuning scheduler.

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Weekly Project Log Report

Week: Week 14 & 15

Date: _____

Task Completed

- Implemented the self-learning adaptation engine.
- High-confidence predictions (above threshold τ) are automatically added to dynamic dataset DSL.
- Designed a daemon for periodic fine-tuning of the classifier with a reduced learning rate.
- Implemented recalculation of metric thresholds from the latest data distributions after each update cycle.
- Added version-controlled model updates to track changes in classifier behaviour over time.

Task in Progress

- Testing self-learning stability: ensuring only high-confidence samples are added to prevent noisy learning.
- Monitoring threshold drift across update cycles.

Task to Begin Next Week

- Conduct full model evaluation on held-out validation set.
- Generate confusion matrices, ROC curves, and per-class metrics.

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Weekly Project Log Report

Week: Week 16

Date: _____

Task Completed

- Conducted comprehensive model evaluation on held-out validation set of unseen drone sequences.
- Achieved peak validation accuracy of approximately 85% with EfficientNet-B0.
- BAD class achieved high precision of 0.8125; RECOVERABLE class achieved top recall of 0.8182.
- Generated normalized confusion matrix revealing that most misclassifications occur between GOOD and RECOVERABLE.
- Validated enhancement results qualitatively: improved contrast, edge sharpness, and feature prominence in RECOVERABLE images.

Task in Progress

- Comparing performance against baseline models: BRISQUE+SVM, VGG16, ResNet50 (Table 9.1 analysis).
- Documenting qualitative enhancement gains for 3D reconstruction metrics.

Task to Begin Next Week

- Finalize model selection and begin system integration.
- Prepare Phase 2 presentation materials.

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Weekly Project Log Report

Week: Week 17

Date: _____

Task Completed

- Integrated all pipeline components: dual-path labeler, EfficientNet-B0 classifier, hybrid enhancement module, self-learning daemon.
- Conducted end-to-end testing on full drone mission sequences.
- Verified that GOOD images pass through without modification and reach photogrammetry pipeline correctly.
- Validated that RECOVERABLE images show measurable improvement after enhancement and successful re-classification.
- Collected and analyzed full evaluation metrics: precision, recall, F1 per class.

Task in Progress

- Calibrating confidence thresholds to optimize self-learning reliability.
- Stress-testing pipeline with high-volume batch inputs.

Task to Begin Next Week

- Finalize model and pipeline; prepare for deployment.
- Begin documentation of final pipeline for report writing.

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Weekly Project Log Report

Week: Week 18 & 19

Date: _____

Task Completed

- Containerized the full pipeline for consistent deployment across environments.
- Configured NVIDIA Jetson deployment path for onboard UAV processing.
- Set up ground station deployment for high-definition post-mission processing.
- Implemented auto-logging and scheduled retraining with rollback fallbacks.
- Validated deployment on sample drone mission data from agricultural and terrain scenarios.

Task in Progress

- Testing onboard inference latency and memory footprint on Jetson hardware.
- Refining rolling update mechanism with version-controlled model weights.

Task to Begin Next Week

- Complete frontend development and API integration.
- Implement PDF report generation for quality assessment outputs.

Suggestions by Guide:

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Weekly Project Log Report

Week: Week 20

Date: _____

Task Completed

- Designed and implemented frontend interface for pipeline interaction.
- Connected backend pipeline components via API for seamless image upload and result retrieval.
- Built visualization pages displaying classification results, enhanced image outputs, and quality score logs.
- Finalized project report, including all chapters, results tables, and figures.
- Completed all documentation and prepared for final project submission.

Task in Progress

- Final testing of frontend-backend integration.
- Reviewing report for completeness and accuracy.

Task to Begin Next Week

- Final calibration and threshold tuning.
- Complete frontend-backend integration.
- Implement PDF summary report generation for final outputs.

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